

Linked List – One Page Memory Map (Interview Ready)

CORE STRUCTURE

Node = data + next

LinkedList = head

INSERT OPERATIONS

Insert at Beginning → new.next = head → head = new

Insert at End → traverse till next == null → last.next = new

DELETE OPERATIONS

Delete First → head = head.next

Delete Last → stop at second last → secondLast.next = null

Delete by Value → skip node → prev.next = curr.next

TRAVERSAL

while (current) → process → current = current.next

SEARCH

Linear scan from head → compare data

REVERSE LINKED LIST (MOST IMPORTANT)

Use 3 pointers: prev, current, next

Steps:

1. next = current.next

2. current.next = prev

3. prev = current

4. current = next

Finally → head = prev

CYCLE DETECTION

Slow pointer moves 1 step

Fast pointer moves 2 steps

If slow == fast → cycle exists

TIME & SPACE COMPLEXITY

Insert / Delete / Search / Reverse → $O(n)$ time

Extra space → $O(1)$

INTERVIEW OPENING LINE

“I’ll first define Node and LinkedList, then implement basic operations step by step.”